Multiplying and Dividing Rational Expressions

Algebra 2 Lesson 4-3 (fast track) Name _____ Date _____ Period ____

Learning goal: I can simplify a rational expression and find the product and the quotient of rational expressions, and state the domain.

Language goal: I can explain how I simplified a rational expression using the words factor, common factor, reciprocal, domain and simplified form.

Spanish: *expresión racional* (rational expression), *máximo factor común* (greatest common factor).

On the quiz: (2) graded: simplify and state the domain. Factor, take the domain from the *original* denominator, divide out common factors (Example 2). (4) graded: divide. Multiply by the reciprocal of the divisor, then factor and divide out (Example 5). (5) bonus: width = area ÷ length (the Practice after Try It 5). (1) simplify and (3) multiply: full Savvas quiz only.

On the Topic 4 test: Question 19 asks for the length of a rectangle from its area and width: factor the area, then divide out the width $x + 6$ (Example 2).

Example 2: Simplify a Rational Expression

What is the simplified form of the rational expression $\dfrac{4 - x^2}{x^2 + 3x - 10}$? What is the domain for which the identity between the two expressions is valid? The **simplified form** has no common factors, other than 1, in the numerator and the denominator.

$$\begin{aligned}
\frac{4 - x^2}{x^2 + 3x - 10} &= \frac{(2 - x)(2 + x)}{(x - 2)(x + 5)} && \textbf{factor} \\
&= \frac{-(x - 2)(x + 2)}{(x - 2)(x + 5)} && 2 - x = -1(x - 2) \\
&= -\frac{x - 2}{x - 2} \cdot \frac{x + 2}{x + 5} && \textbf{common factor} \\
&= -\frac{x + 2}{x + 5} && \text{divides to 1}
\end{aligned}$$

Simplified form: $-\dfrac{x + 2}{x + 5}$ for $x \neq 2$ or -5

Check $x = 0$: original $\dfrac{4 - 0}{0 + 0 - 10} = -\dfrac{2}{5}$; simplified $-\dfrac{0 + 2}{0 + 5} = -\dfrac{2}{5}$ ✓

Watch out: only *factors* divide out, never terms. You cannot cross out the x^2 terms.

Domain (from the *original* denominator): $(x-2)(x+5) = 0$ when $x = 2$ or $x = -5$, so exclude both.

Say it: I _____ the top and the bottom. The common factor _____ divides to 1. The domain leaves out $x =$ _____ and $x =$ _____ because they make the _____ equal zero.

Try It 2: Simplify each expression and state the domain

a. $\dfrac{x^2 + 2x + 1}{x^3 - 2x^2 - 3x}$

Simplified: _____ Domain: $x \neq$ _____

b. $\dfrac{x^3 + 4x^2 - x - 4}{x^2 + 3x - 4}$

Hint: factor the numerator by grouping.

Simplified: _____ Domain: $x \neq$ _____

Example 5: Divide Rational Expressions

What is the quotient of $\dfrac{x^3+3x^2+3x+1}{1-x^2}$ and $\dfrac{x^2+5x+4}{x^2+3x-4}$?

The idea: divide fractions the way you always have. Multiply by the **reciprocal** of the divisor (flip the second fraction), then it is a multiplication problem.

1. Multiply by the **reciprocal** of the divisor.

$$\frac{x^3+3x^2+3x+1}{1-x^2} \div \frac{x^2+5x+4}{x^2+3x-4} = \frac{x^3+3x^2+3x+1}{1-x^2} \cdot \frac{x^2+3x-4}{x^2+5x+4}$$

2. **Factor** every polynomial.

$x^3+3x^2+3x+1 = (x+1)(x+1)(x+1)$ (a perfect cube: $(x+1)^3$)

$1-x^2 = (1-x)(1+x) = -(x-1)(x+1)$ (opposites again, like Example 2)

$x^2+3x-4 = (x+4)(x-1)$ $\qquad$ $x^2+5x+4 = (x+1)(x+4)$

3. **Domain:** leave out every x that makes a denominator zero, *including the divisor's numerator*, because it moved to the bottom.
$1-x^2=0$: $x=\pm 1$; $\quad x^2+3x-4=0$: $x=-4,\ 1$; $\quad x^2+5x+4=0$: $x=-1,\ -4$. So $x \neq -4,\ -1,\ 1$.

4. Divide out the **common factors**.

$$= \frac{(x+1)(x+1)(x+1)(x+4)(x-1)}{-(x-1)(x+1)(x+1)(x+4)} = \frac{x+1}{-1} \cdot \frac{x+1}{x+1} \cdot \frac{x+1}{x+1} \cdot \frac{x-1}{x-1} \cdot \frac{x+4}{x+4}$$

The quotient is $-(x+1),\ x \neq -4,\ -1,$ or 1.

Check with $x=0$: $\dfrac{1}{1} \div \dfrac{4}{-4} = 1 \div (-1) = -1$, and $-(0+1) = -1$ ✓

Say it: To divide, I multiplied by the _____ of the divisor. Then I _____ every polynomial and divided out the _____. The domain leaves out $x =$ _____ because

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Try It 5: Find the simplified quotient and the domain of each expression

a. $\dfrac{1}{x^2+9x} \div \left(\dfrac{6-x}{3x^2-18x}\right)$

Quotient: _____

Domain: all x except _____

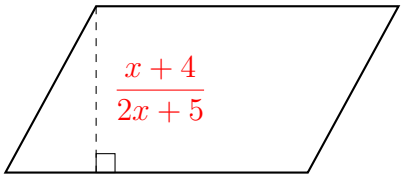
b. $\dfrac{2x^2-12x}{x+5} \div \left(\dfrac{x-6}{x+5}\right)$

Quotient: _____

Domain: all x except _____

Practice: Area divided by one dimension (Practice 37)

A parallelogram with an area of $\frac{3x+12}{10x+25}$ square units has a height shown. Find the length of the base of the parallelogram.



a. Area = base × height, so base = area ÷ _____.
b. Multiply by the reciprocal, factor, and divide out the common factors.

c. Base = _____ units.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
37N ★	Simplify rational expressions	Quiz 2; Quiz 1 (full Savvas quiz only)
79Q ★	Domain of rational functions	Quiz 2: the domain (Example 2, step 2)
MG2 ★	Multiply and divide rational expressions	Quiz 4, 5; Quiz 3 (full Savvas quiz only)
UB5	Factor quadratics	Warm-up: every problem starts by factoring
QKF	Factor using a quadratic pattern	Warm-up: $4-x^2$, $9-x^2$, x^2+6x+9

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Practice 13 (DOK 3)

Why are rational expressions closed under multiplication and division?

Start from two rational expressions $\frac{a}{b}$ and $\frac{c}{d}$, where a, b, c, d are polynomials and $b, d \neq 0$.

$\frac{a}{b} \cdot \frac{c}{d} =$ _____. The top and the bottom are both _____, so the product is a _____.

$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot$ _____ because dividing means multiplying by the _____. So the quotient is also a _____ (as long as $c \neq 0$).

Before the quiz, finish this sentence. To simplify a product or quotient, I first _____ every polynomial. The _____ leaves out every x that makes an original denominator zero or, in a quotient, the divisor's numerator zero. Last I divide out each _____.